Two-channel signatures of the E_7 spectrum in a blockaded Rydberg chain

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Uniform and staggered detuning probes place odd and even excitations on a common spectral scale in Rydberg chains near tricritical Ising criticality. In a strictly blockaded model with long-range interactions, we find two candidate peaks per channel whose energy ratios lie within 3% of the four lightest E_7 particles in a direction-dependent parameter region. Their spectral-weight trends support this assignment, but the shallow A_4 candidate retains a binding about 62% above the ideal value relative to the lightest excitation at the working point. Changing the interaction at fixed lowest gap reveals additional size dependence, preventing a controlled test of the predicted universal response. On open chains, even length forbids same-state leakage between symmetric probes by reflection symmetry; odd length does not. The heavy particles A_5–A_7 remain unresolved. Together, energy ratios, spectral weights, and probe symmetry provide a finite-chain benchmark for two-channel E_7 spectroscopy.

Introduction.—Near a quantum critical point, a microscopic chain can support collective excitations governed by a continuum field theory [1]. Certain perturbations produce particle spectra fixed by exceptional Lie algebras. The magnetic perturbation of critical Ising theory realizes E_8 [2], with experimental signatures in $CoNb_2O_6$ [3] and Rydberg arrays [4]. The leading thermal perturbation of the tricritical Ising (TCI) theory instead produces seven particles with the E_7 mass spectrum [5–7]. In its disordered phase, these excitations separate into two internal symmetry sectors [8], suggesting that complementary probes can reveal more of the spectrum than a single drive.

Individually controlled Rydberg atoms provide a setting for such spectroscopy [9, 10]. Experiments have explored Ising critical dynamics [11, 12], while blockaded chains connect to Fibonacci anyon models [13] and support tricriticality under effective attraction [14]. Sun *et al.* [15] reached TCI by preparing the upper edge of the blockaded sector and measured finite-size critical spectra. Within the projected model, reversing the Hamiltonian turns this into an attractive ground-state problem. Its neighboring gapped regime is therefore a natural place to seek the E_7 excitations.

The relation between atomic density and continuum fields [14], the relevant form factors [7, 16, 17], and the sensitivity of the shallow A_4 bound state [8, 16] are established. Király and Lencsés [18] obtained Blume–Capel quench spectra compatible with the even particles A_2, A_4, A_5, including form-factor amplitude comparisons; their mass scale is calibrated with A_2. Here uniform and staggered density probes place odd and even light-particle candidates on a common spectral scale in a Rydberg chain. We map their finite-chain matching region and combine energy ratios, spectral weights, and reflection selection with binding-energy and fixed-gap-response tests of convergence.

Two complementary probes.—We consider a driven chain in which neighboring atoms cannot both occupy the Rydberg state. This constraint connects hard-boson models [19] and blockaded-spin descriptions [20]. The laser has Rabi frequency Ω and detuning Δ; the effective attraction is specified by its next-nearest-neighbor strength V_2. All allowed pairs interact with an inverse-sixth-power tail. Energies are measured in units of Ω. This strict-blockade model omits finite-blockade corrections to the experimental Hamiltonian [15]; open chains have no additional boundary detuning.

A uniform detuning drive changes all site energies together and couples principally to the continuum energy density. A staggered drive alternates its sign between neighboring atoms and couples principally to the order parameter distinguishing the two density-wave patterns [14]. The staggered coupling was proposed in Ref. [15]. In the periodic bulk, the former addresses even excitations, including A_2 and A_4, and the latter odd excitations, including A_1 and A_3. The measured object is the weak-drive density response [21]. Symmetry allows these peaks but does not ensure appreciable weight or separation from other excitations. The model, field correspondence, response definitions, and numerical methods are given in Supplemental Material (SM), Secs. S1–S2.

Four candidate peaks.—We calculate ground-state responses for periodic chains up to 36 atoms and open chains of 19–28 atoms. Figure 1 shows a direction-dependent region in which the three light-particle energy ratios agree with E_7 within 3%. At the reference detuning -1.5695Ω and attraction -1.4381Ω, the lightest gap is 0.9196Ω. The next three candidate ratios are 1.2678, 1.9243, and 1.9508, differing from E_7 by -1.39%, $+2.39\%$, and -0.95%, respectively. These are finite-chain assignments based on energy ordering and probe symmetry, rather than independently identified continuum particles.

The periodic spectrum contains two prominent subthreshold candidates per channel [Fig. 2(a)]. The corresponding open-chain response retains four dominant lines with ratios within 2% of E_7 [Fig. 2(b)]. Agreement at a selected size does not establish convergence: closer to the estimated critical point, one candidate ratio still changes substantially between 28 and 36 atoms. Access to the continuum regime requires increasing the chain length as

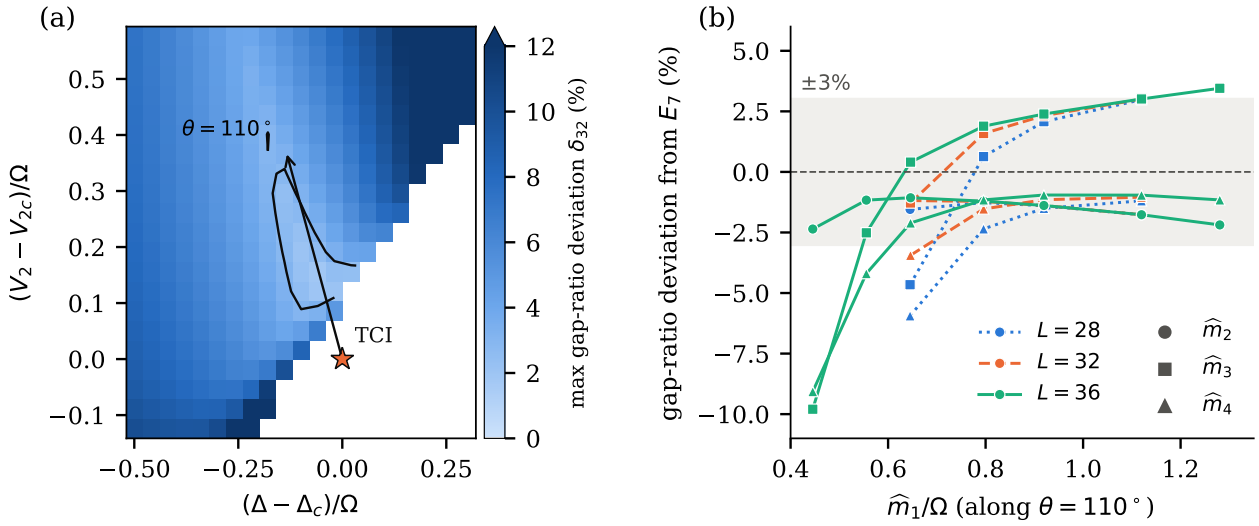


FIG. 1. Finite-chain spectral matching. (a) Largest relative deviation of the three candidate energy ratios from E_7, in percent, for 32 atoms. The contour marks 3%. The star estimates TCI; the arrow marks the direction $\theta = 110^\circ$. Blank cells fail the numerical screening described in Supplemental Material (SM), Sec. S3; they do not delimit a phase. (b) Individual ratio deviations along this direction. Symbols label candidate levels and line styles label sizes; the shaded band is ±3%. Lines guide the eye. This map is not an experimental error budget.

the gap is reduced. SM, Sec. S3, specifies the scan and its size dependence.

A shallow bound state tests the assignment.—Binding energy separates mass-ratio agreement from convergence for A_4. In the ideal theory it is a weakly bound pair of A_1 particles: its binding is only 3.04% of the lightest mass. A small error in its excitation energy can therefore hide a much larger error in the binding. Our binding decreases from 15.9% of the lowest gap at 18 atoms to 4.92% at 36 atoms [Fig. 3(a)]. The latter remains about 62% above the ideal value, despite the subpercent error in the candidate mass ratio.

A loosely bound pair is spatially extended and can sense the finite chain well beyond the ordinary correlation length [22, 23]. Using the ideal bound-state relation and the critical velocity gives a diagnostic extent of roughly 12 sites, compared with an ordinary correlation length of roughly two sites. This is an estimate, not a measured velocity in the gapped regime. Fits over successive size windows drift; they do not separate finite-volume effects from lattice corrections or establish a controlled infinite-chain binding (SM, Sec. S3).

Spectral weights offer additional support [24]. Between 24 and 36 atoms, the A_1 and A_2 weights change by less than 0.4%, while A_3 gains 9.6% and A_4 loses 11.2%. The next levels in both sectors lose weight more rapidly [Fig. 3(b)]. These trends are compatible with light particles and a discretized multiparticle background, but finite-size effects and the mixing of continuum fields in the density probe remain unresolved.

Changing the interaction at fixed gap.—A stronger test asks whether the candidate energies respond to a perturbation in the proportions expected from the continuum theory [8, 25]. The known subleading thermal perturbation initially shifts all three mass ratios in the same direction. It cannot by itself explain the opposite signs of the second- and third-particle deviations at our reference point. This does not rule out an eventual approach to E_7 on larger chains.

We vary the interaction and retune the detuning to keep the lightest gap fixed, testing changes in spectral shape without changing that reference energy. We compare 32 and 36 atoms at lowest gaps of 0.75Ω and 0.90Ω. At the larger gap, the two relative responses change by 0.89% and 4.19% between sizes; at the smaller gap they change by 5.40% and 15.26%. Neither gap meets the preselected 3% size-stability criterion for both responses. Numerical cross-checks are much more precise than these drifts. Lowering the gap worsens both the size dependence and agreement with the continuum prediction. The definitions and all four comparisons appear in SM, Sec. S4 and Table S1. The present chains therefore do not establish a universal perturbative response, even though one response ratio is close to its predicted value. Keeping the gap fixed also leaves other properties, including the velocity, free to change.

Boundaries and spectral backgrounds.—Open chains replace momentum selection with spatial reflection, which differs from the internal symmetry of the continuum theory. For an even number of atoms, symmetric uniform and staggered drives have opposite reflection

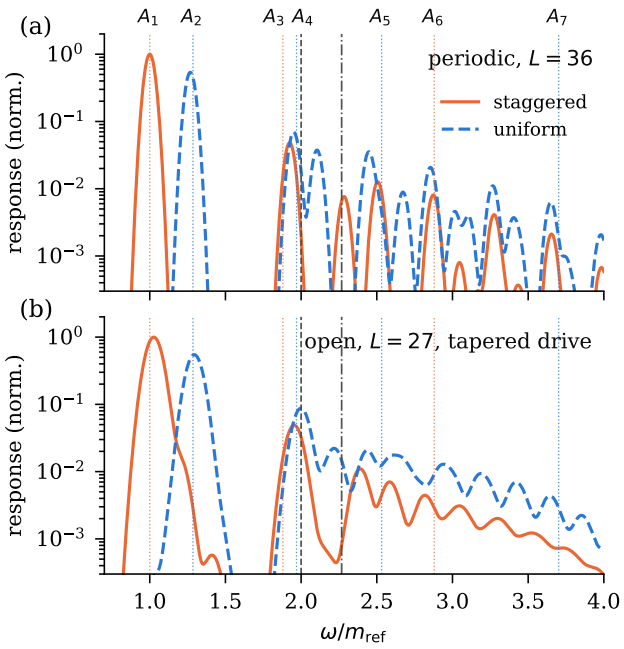


FIG. 2. Response at the reference point: (a) periodic 36-atom chain; (b) open 27-atom chain with a smooth drive that weakens near both ends. Energies are scaled by the periodic lowest gap, $m_{\mathrm{ref}} = 0.9196\Omega$; each panel is normalized to its own staggered peak height. Solid orange: staggered; dashed blue: uniform. Only (a) has exact momentum selection. Gaussian widths of $0.03m_{\mathrm{ref}}$ and $0.05m_{\mathrm{ref}}$ illustrate smoothing, not instrumental linewidths. Dotted lines and particle labels mark ideal E_7 ratios; only A_1–A_4 are assigned candidate peaks. Gray lines mark the lowest two-particle gap sums from the periodic chain, not measured open-chain continuum edges.

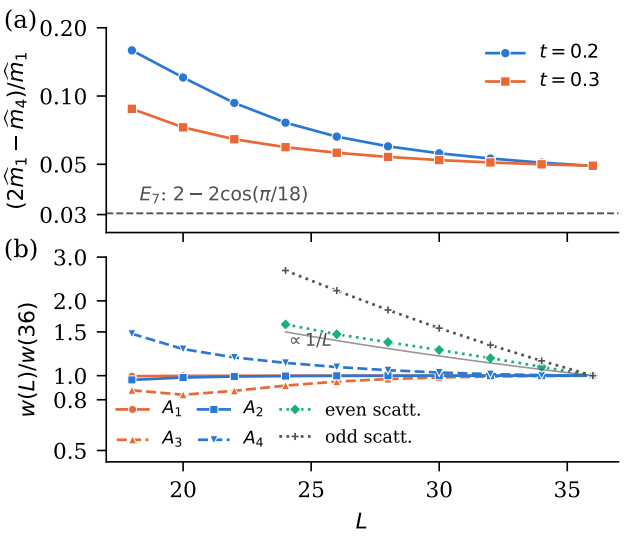


FIG. 3. Size dependence along $\theta = 110°$. (a) Binding relative to the lowest gap at two distances from TCI: $t = 0.2$ is the reference point; $t = 0.3$ is farther away. The dashed line marks the ideal E_7 binding. (b) Each spectral weight is divided by its own 36-atom value. Orange: odd sector; blue: even sector. The four particle-labelled curves are candidates: A_1 and A_2 are nearly size independent, while A_3 and A_4 retain finite-size variation. Dotted curves are the next levels in each sector. The gray inverse-length curve is a reference, not a fit. The common endpoint at unity is imposed by normalization. Definitions are in SM, Sec. S3.

parity. From the same ground state they cannot excite the same reflection eigenstate. For an odd number, both drives are reflection even, so they can share spectral lines. This distinction follows from reflecting the drive pattern; the derivation and complete leakage table are in SM, Sec. S5 and Table S2.

For the tapered odd chains, the largest unintended-to-intended weight ratio falls from 39.4% at 19 atoms to 7.46% at 27 atoms. Tapering is not uniformly beneficial: at 27 atoms it suppresses leakage into the A_1 candidate but slightly increases leakage into A_4. Nearby lines also remain within each channel. A staggered satellite is compatible with a standing-wave excitation, although a boundary origin is not excluded. Thus, exact same-state selection on even chains does not ensure resolved peaks once finite linewidths are included.

The lowest even-channel continuum contains two A_1 particles; the lowest odd-channel continuum contains an A_1 and an A_2. Their onsets differ because the identical-particle matrix element vanishes at threshold, while the mixed-particle one does not [16]. Our next discrete levels move toward these gap sums with increasing size, but do not verify the predicted onset power laws (SM, Sec. S6). In particular, the first even level above the A_4 candidate still lies above the continuum estimate, exaggerating its apparent separation.

We do not reliably resolve A_5, A_6, or A_7 from combined energy and weight trends. Their ideal energies lie above symmetry-allowed continua. Integrability-breaking decays can be studied through perturbation theory [26], with the relevant E_7 channels discussed in Ref. [8]. Finite closed chains do not measure decay widths; our unresolved levels neither establish nor exclude heavy resonances in larger systems.

Experimental implications.—An experiment would prepare the gapped regime next to the demonstrated TCI point and compare weak uniform and staggered responses. The two-wavelength tweezer arrangement proposed in Ref. [15] offers one route to a sign-changing drive; it has not yet been demonstrated in that setup. Even atom numbers provide a useful reflection-based separation of states, while tapering changes their weights.

The shallow binding sets a demanding resolution scale. At a Rabi frequency of 5.5 MHz, the finite-chain binding corresponds to about 0.25 MHz; the ideal E_7 value at the same lowest gap is about 0.15 MHz. The illustrative open-chain smoothing in Fig. 2 has a full width at half maximum of about 0.60 MHz, so the displayed

curves do not demonstrate direct resolution of this binding. No signal-to-noise or joint line-fitting analysis has been performed.

As a single calibration check, changing the interaction by $\pm 1.7\%$ at fixed detuning changes the second-, third-, and fourth-particle errors relative to E_7 by at most 0.13, 0.43, and 0.20 percentage points. The lowest gap changes by up to 4.5%. This is not a complete error budget: finite-blockade corrections, preparation, decoherence, and readout remain unmodeled. Detailed definitions and conversions are in SM, Sec. S6.

Conclusion.—Uniform and staggered density probes combine energy, weight, and symmetry information in a common finite-chain test of the four lightest E_7 candidates. Even open chains provide exact same-state probe separation through reflection symmetry. The shallow binding and fixed-gap response test convergence beyond mass-ratio agreement: finite-size and lattice corrections remain inseparable at the studied lengths. These results define a two-channel spectroscopy benchmark; controlled particle identification, experimental resolvability, and the heavy sector remain open.

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